IT Project Management

Lecture 05
Project Scope Management (chapter 5 in book)

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Learning Objectives

- Understand the importance of good project scope management
- Describe the process of planning scope management
- Discuss methods for collecting and documenting requirements to meet stakeholder needs and expectations
- Explain the scope definition process and describe the contents of a project scope statement
- Discuss the process for creating a work breakdown structure using the analogy,

Learning Objectives

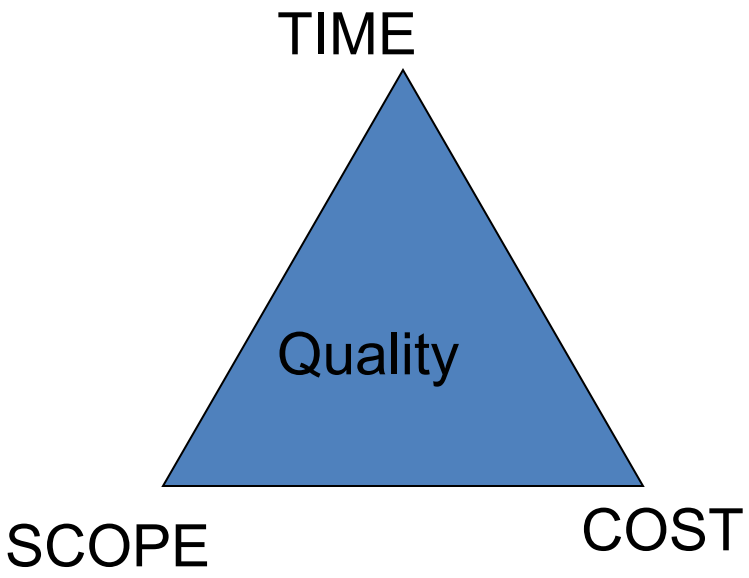
- Explain the importance of validating scope and how it relates to defining and controlling scope
- Understand the importance of controlling scope and approaches for preventing scope-related problems on information technology (IT) projects
- Describe how software can assist in project scope management

What is Project Scope Management?

- **Scope** refers to *all* the work involved in creating the products of the project and the processes used to create them
- A **deliverable** is a product produced as part of a project, such as hardware or software, planning documents, or meeting minutes
- Project scope management includes the processes involved in defining and controlling what is or is not included in a project

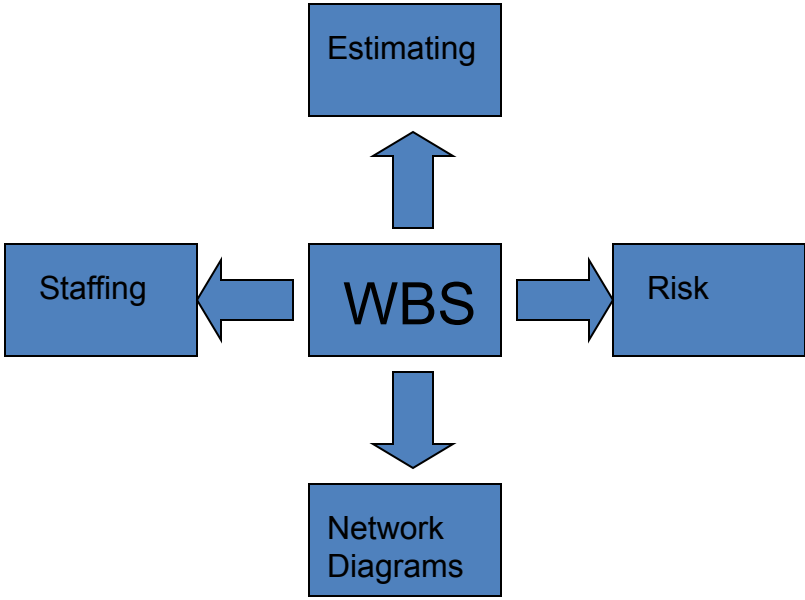
Why Do We Manage Scope?

- Can’t manage schedule and budget if scope is out of control (Triple Constraint)
- Scope docs are used to manage expectations



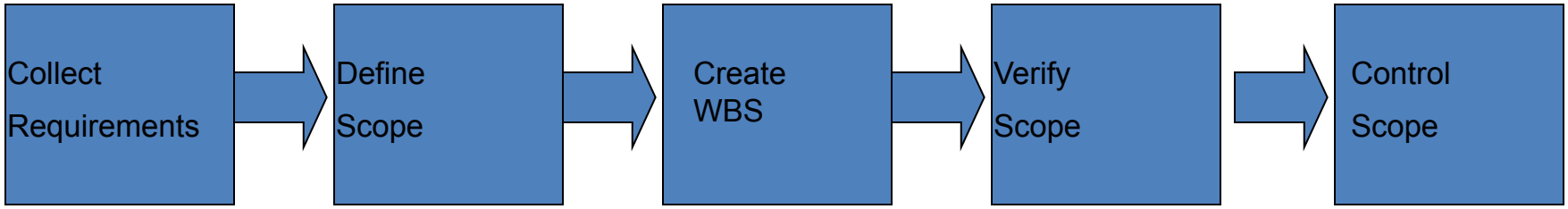
Scope Management Key Points

- What is scope management
  - Checking to ensure that one is completing work
  - Saying No to additional work not in the charter
  - Preventing extra work/gold plating
- Work Breakdown Structure (WBS)
  - Foundation of the project, all planning and controlling is based on the WBS
  - Identifies all work to be performed, if it is not in the WBS it does not need to be done
  - Graphical picture of work

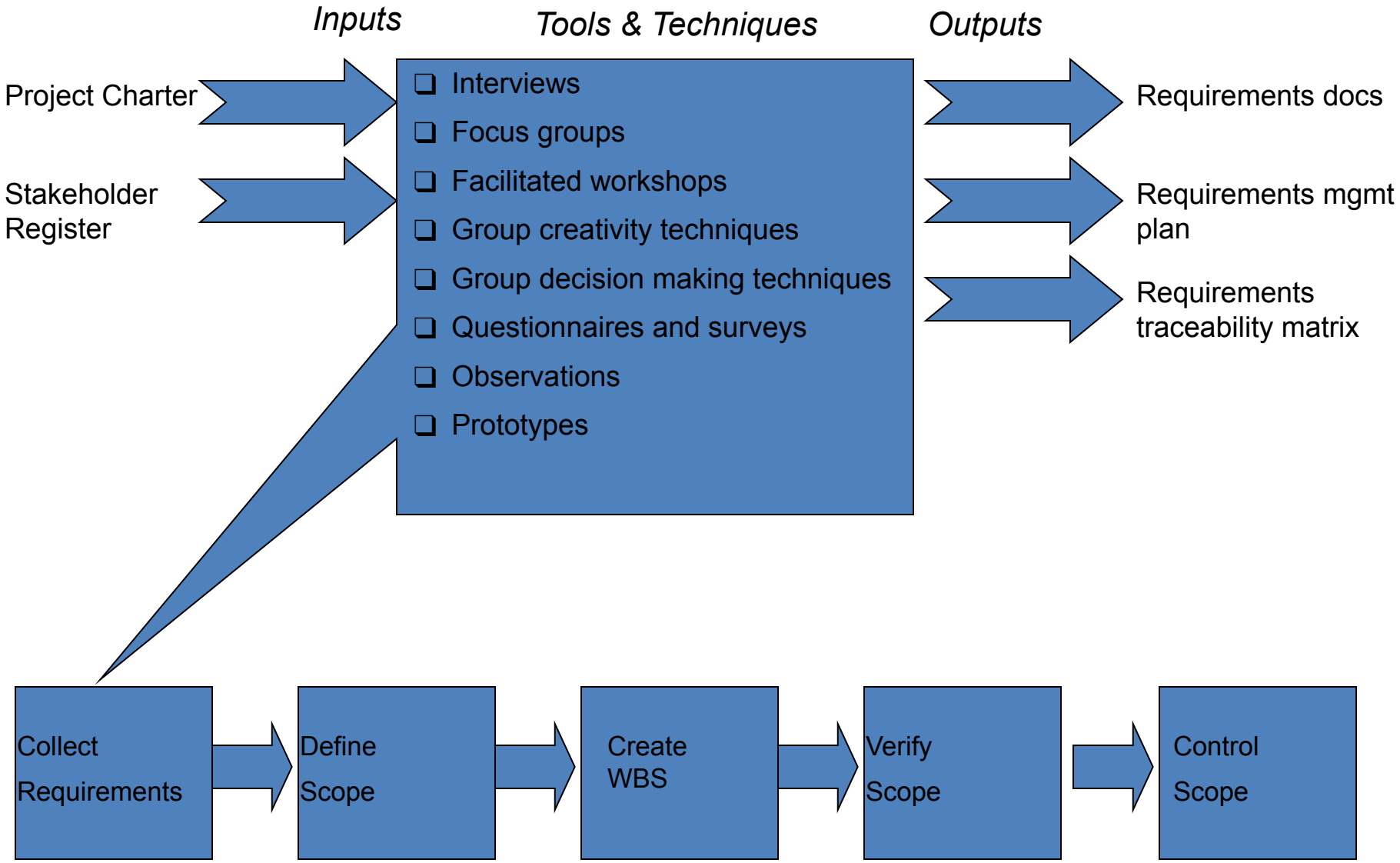


How Do We Manage Scope?

- Five processes
  - Collect Requirements
  - Define Scope
  - Create WBS
  - Verify Scope
  - Control Scope



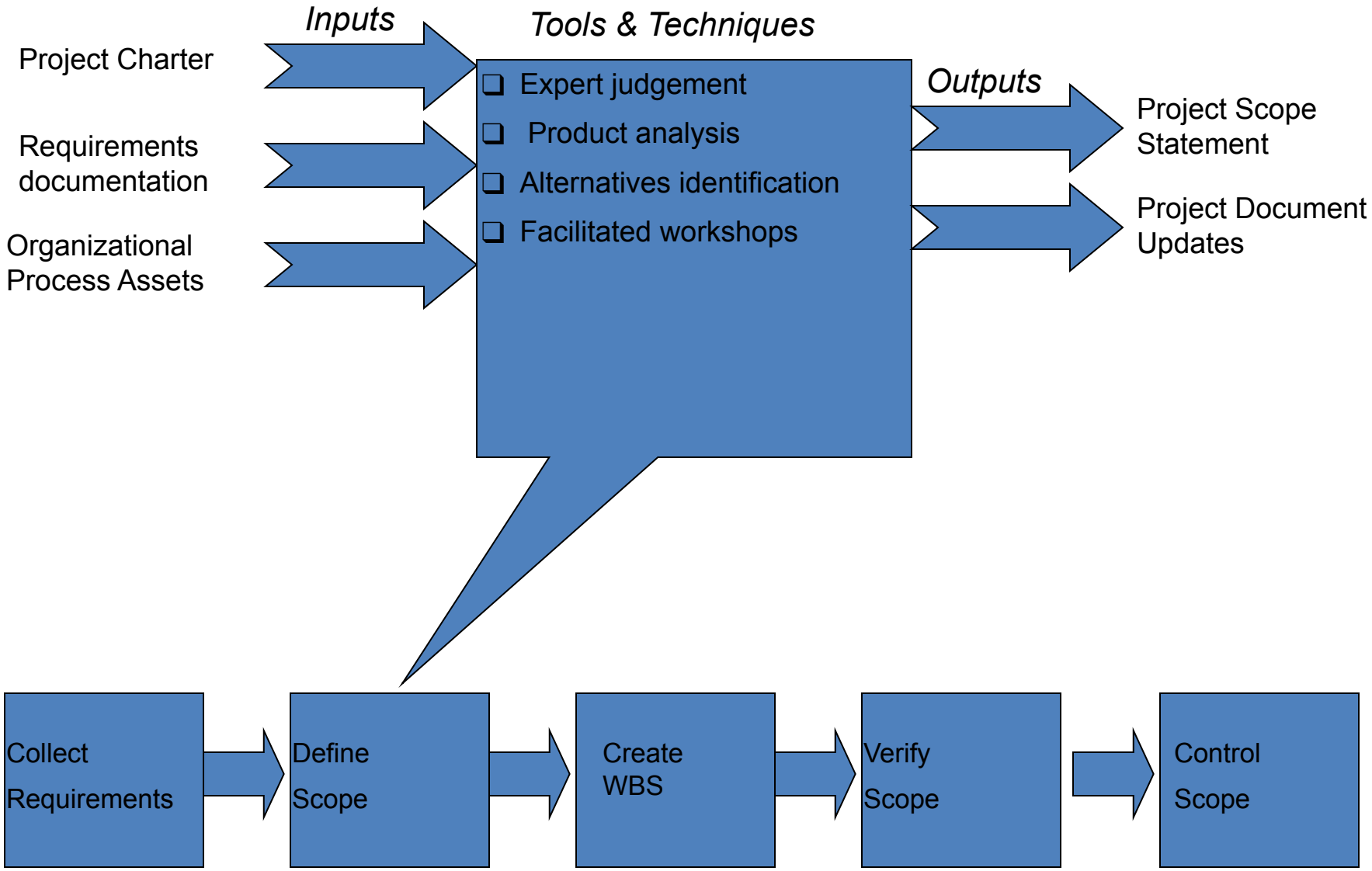
Collect Requirements



Collect Requirements

- Determine product of the project requirements
- Make sure all requirements support the business need of the project as described in the charter

Define Scope

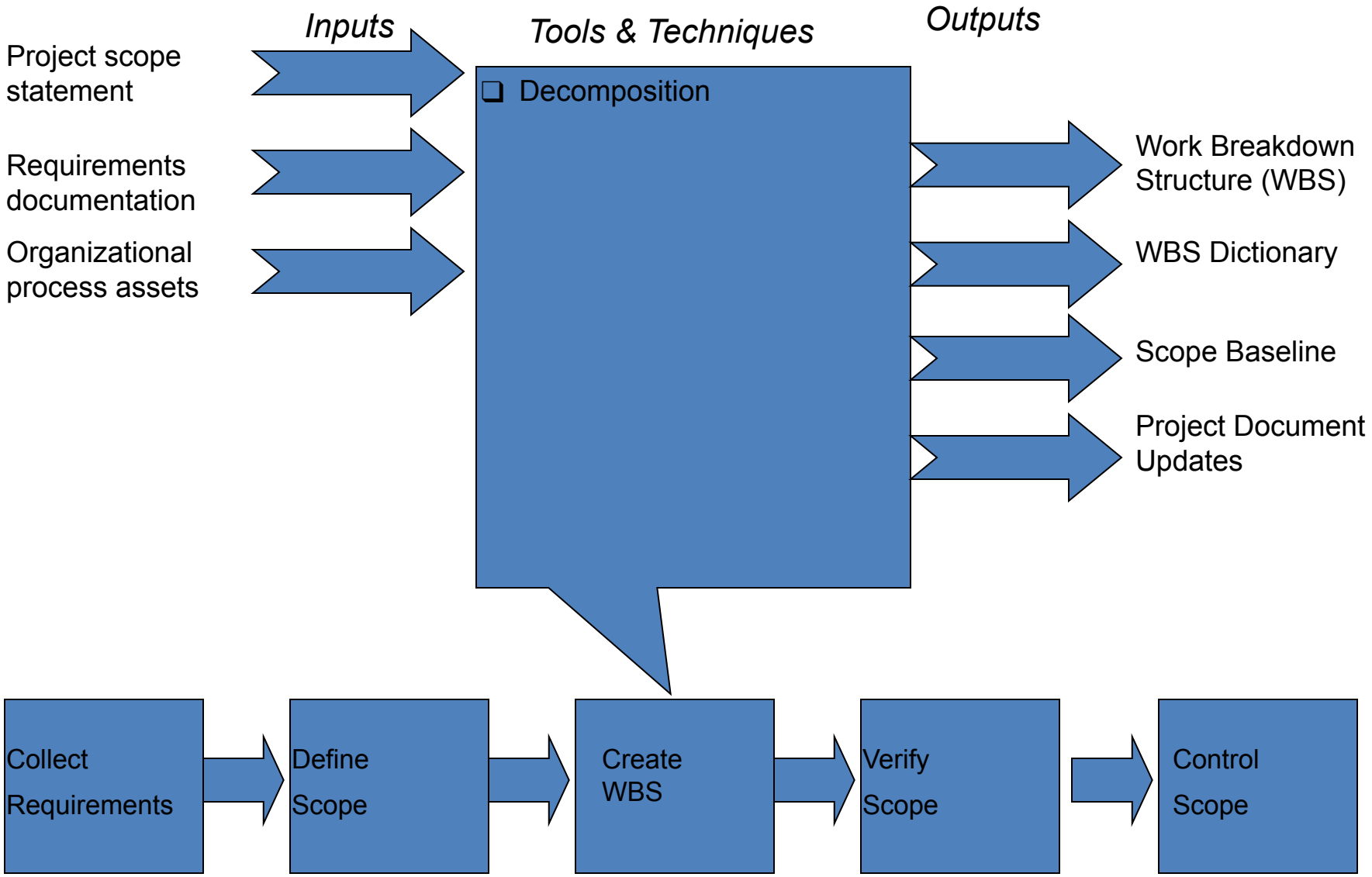


Project Scope Statement

- Product scope description
- Product acceptance criteria
- Project deliverables
- Project exclusions
- Project constraints
- Project assumptions

Remember the difference between product and project?

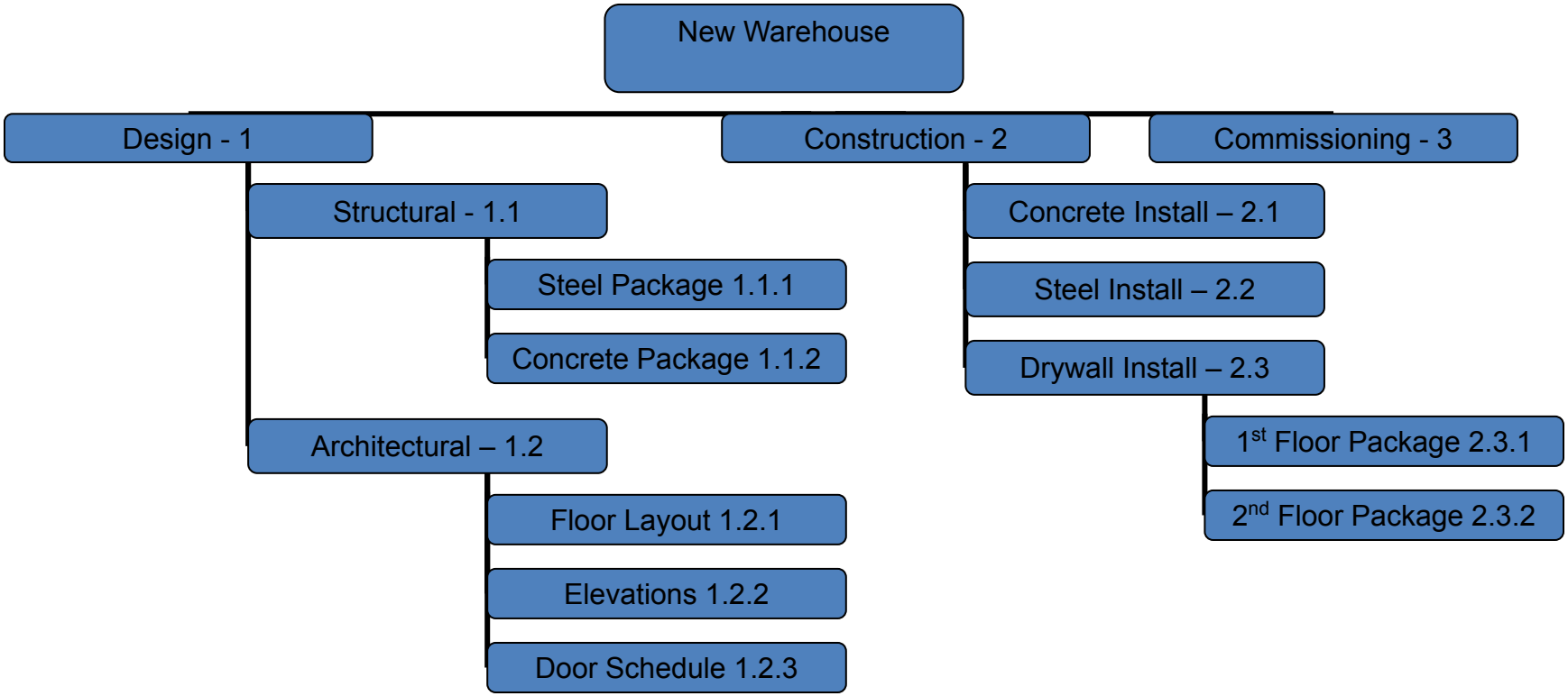
Create WBS


Inputs
Tools & Techniques
Outputs
Project scope statement
Requirements documentation
Organizational process assets
❑ Decomposition
Work Breakdown Structure (WBS)
WBS Dictionary
Scope Baseline
Project Document Updates
Collect Requirements
Define Scope
Create WBS
Verify Scope
Control Scope

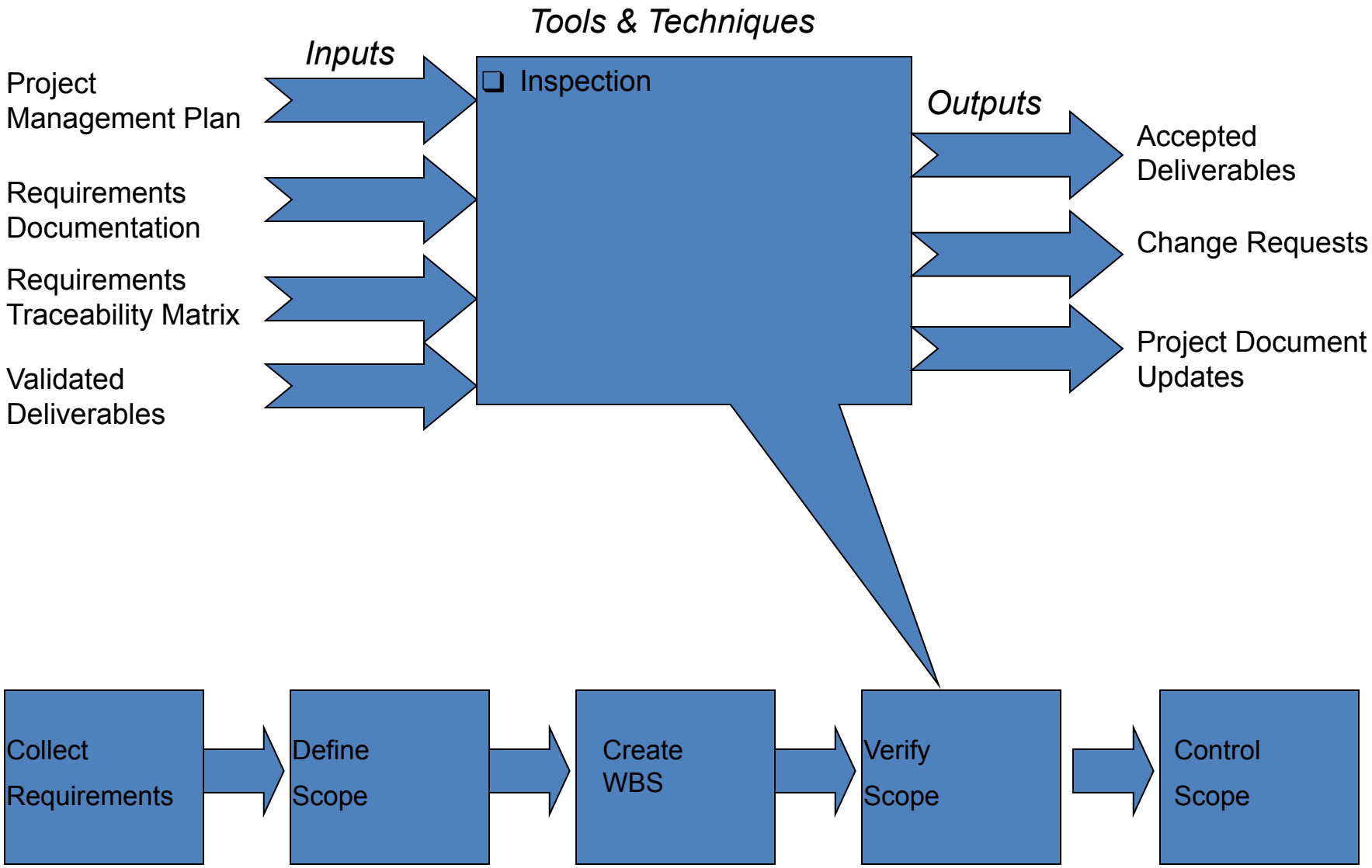
Work Breakdown Structure

- Start with major project deliverables or phases
- “Decomposition” is breaking down the deliverables into more manageable parts
  - Not all branches need the same level of decomposition!
- Decompose into “Work Packages”
  - 8-80 Hour Rule ⇒ Don’t break work down too much

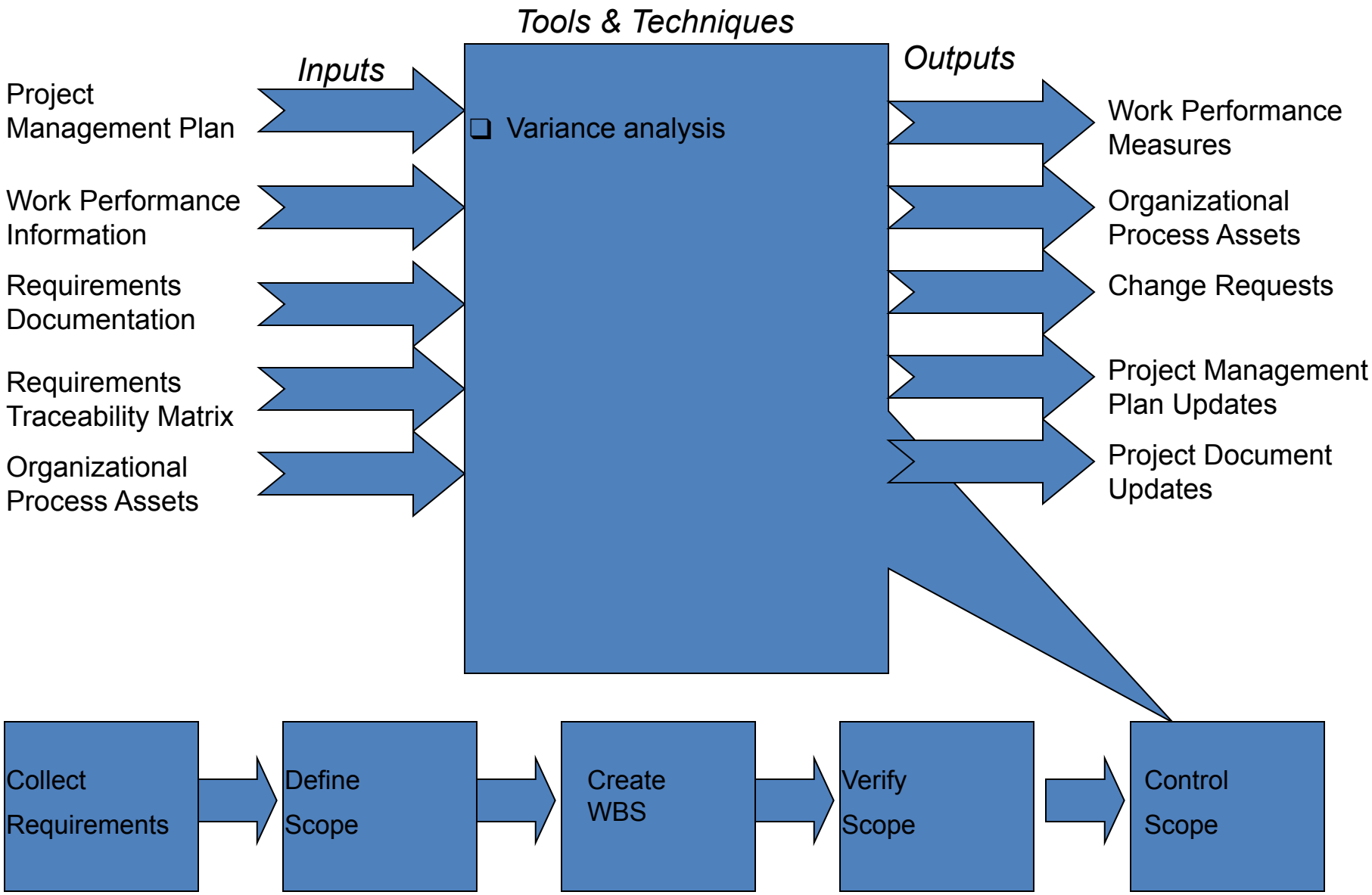
Work Breakdown Structure



Verify Scope



Control Scope



Scope Definitions

- Change Control System
  - Documentation, tracking, and approvals required to change ***project*** scope
- Configuration Management System
  - Documentation, tracking, and approvals required to change ***product*** scope
- Management by Objectives (MBO)
  - Establish unambiguous and realistic objectives
  - Periodically evaluate if objectives are being met
  - Take corrective action

Other Topics – Project Selection

- Benefit Measurement Methods (Comparative Approach)
  - Murder Board – Panel that tries to shoot down new ideas
  - Scoring Models
  - Benefit Compared to Cost
- Constrained Optimization Methods (Mathematical)
  - Linear Programming
  - Integer Programming
  - Multi-Objective Programming

Project Scope Management Processes

- **Planning scope:** determining how the project's scope
- and requirements will be managed
- **Collecting requirements:** defining and documenting the features and functions of the products produced during the project as well as the processes used for creating them
- **Defining scope:** reviewing the project charter, requirements documents, and organizational process assets to create a scope statement
- **Creating the WBS:** subdividing the major project deliverables into smaller, more manageable components
- **Validating scope**: formalizing acceptance of the project deliverables
- **Controlling scope:** controlling changes to project scope throughout the life of the project

Figure 5-1. Project Scope Management Summary

Planning
Process: **Plan scope management**
Outputs: Scope management plan, requirements management plan
Process: **Collect requirements**
Outputs: Requirements documentation, requirements traceability matrix
Process: **Define scope**
Outputs: Project scope statement, project documents updates
Process: **Create WBS**
Outputs: Scope baseline, project documents updates

Monitoring and Controlling
Process: **Validate scope**
Outputs: Accepted deliverables, change requests, work performance information, project documents updates
Process: **Control scope**
Outputs: Work performance information, change requests, project management plan updates, project documents updates, organizational process assets updates

Project Start → **Project Finish**

Planning Scope Management

- The project team uses expert judgment and meetings to develop two important outputs: the scope management plan and the requirements management plan
- The scope management plan is a subsidiary part of the project management plan

Scope Management Plan Contents

- How to prepare a detailed project scope statement
- How to create a WBS
- How to maintain and approve the WBS
- How to obtain formal acceptance of the completed project deliverables
- How to control requests for changes to the project scope

Requirements Management Plan

- The PMBOK® Guide, Fifth Edition, describes requirements as "conditions or capabilities that must be met by the project or present in the product, service, or result to satisfy an agreement or other formally imposed specification"
- The **requirements management plan** documents how project requirements will be analyzed, documented, and managed

Table 5-2. Sample Project Charter (partial)

Project Title: Information Technology (IT) Upgrade Project

Project Start Date: March 4 **Projected Finish Date:** December 4

Key Schedule Milestones:

- Inventory update completed April 15
- Hardware and software acquired August 1
- Installation completed October 1
- Testing completed November 15

Budget Information: Budgeted $1,000,000 for hardware and software costs and $500,000 for labor costs.

Project Manager: Kim Nguyen, (310) 555–2784, knguyen@course.com

Project Objectives: Upgrade hardware and software for all employees (approximately 2,000) within nine months based on new corporate standards. See attached sheet describing the new standards. Upgrades may affect servers as well as associated network hardware and software.

Main Project Success Criteria: The hardware, software, and network upgrades must meet all written specifications, be thoroughly tested, and be completed in nine months. Employee work disruptions will be minimal.

Approach:

- Update the IT inventory database to determine upgrade needs
- Develop detailed cost estimate for project and report to CIO
- Issue a request for quote to obtain hardware and software
- Use internal staff as much as possible for planning, analysis, and installation

Table 5-3: Further Defining Project Scope

Project Charter: Upgrades may affect servers . . . (listed under Project Objectives)
Project Scope Statement, Version 1: Servers: If additional servers are required to support this project, they must be compatible with existing servers. If it is more economical to enhance existing servers, a detailed description of enhancements must be submitted to the CIO for approval. See current server specifications provided in Attachment 6. The CEO must approve a detailed plan describing the servers and their location at least two weeks before installation.
Project Scope Statement, Version 2: Servers: This project will require purchasing ten new servers to support Web, network, database, application, and printing functions. Virtualization will be used to maximize efficiency. Detailed descriptions of the servers are provided in a product brochure in Appendix 8 along with a plan describing where they will be located.

Creating the Work Breakdown Structure (WBS)

- A **WBS** is a deliverable-oriented grouping of the work involved in a project that defines the total scope of the project
- WBS is a foundation document that provides the basis for planning and managing project schedules, costs, resources, and changes
- **Decomposition** is subdividing project deliverables into smaller pieces
- A **work package** is a task at the lowest level of the WBS
- The **scope baseline** includes the approved project

Figure 5-3. Sample Intranet WBS Organized by Product

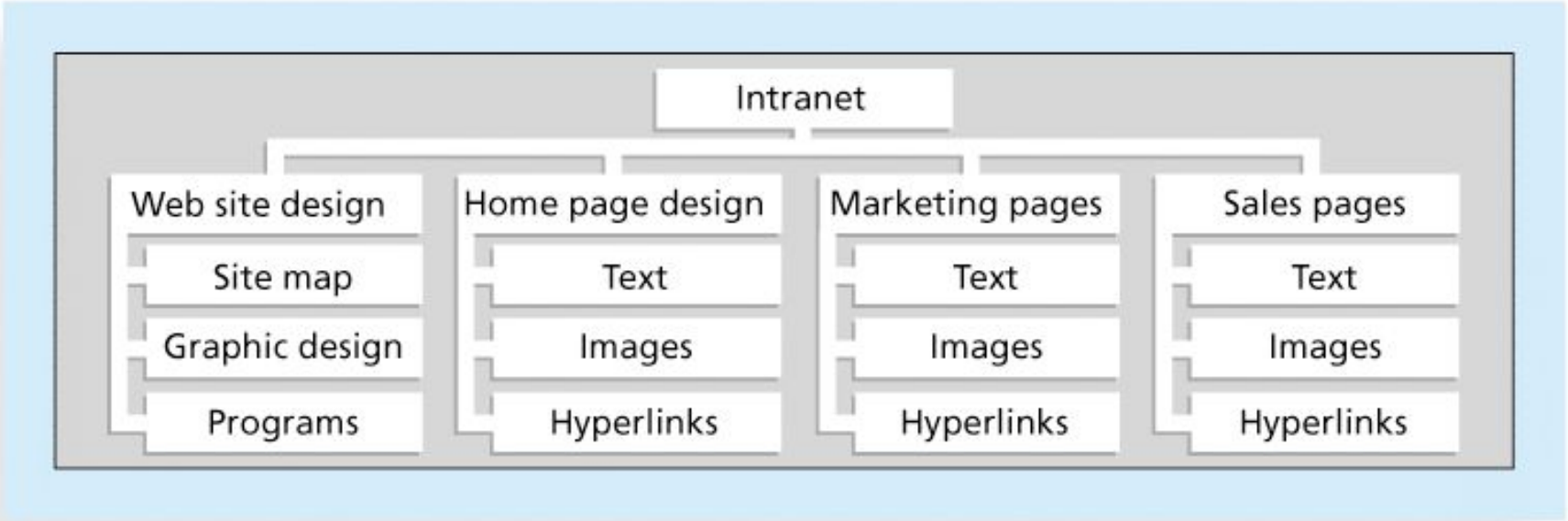
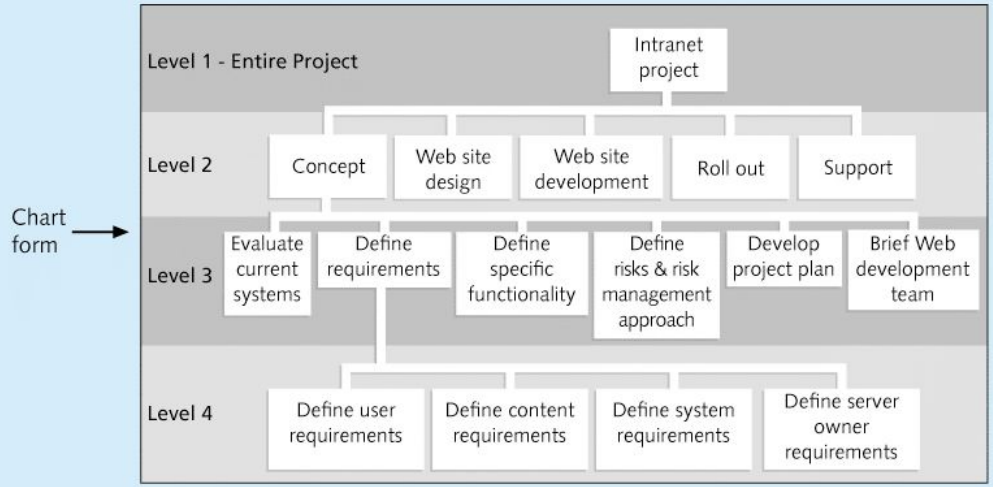


Figure 5-4. Sample Intranet WBS Organized by Phase



Tabular form with PMI numbering

1.1 Concept
- 1.1.1 Evaluate current systems
- 1.1.2 Define requirements
  - 1.1.2.1 Define user requirements
  - 1.1.2.2 Define content requirements
  - 1.1.2.3 Define system requirements
  - 1.1.2.4 Define server owner requirements
- 1.1.3 Define specific functionality
- 1.1.4 Define risks and risk management approach
- 1.1.5 Develop project plan
- 1.1.6 Brief Web development team

1.2 Web site design
1.3 Web site development
1.4 Roll out
1.5 Support

Figure 5-5. Intranet WBS and Gantt Chart in Microsoft Project

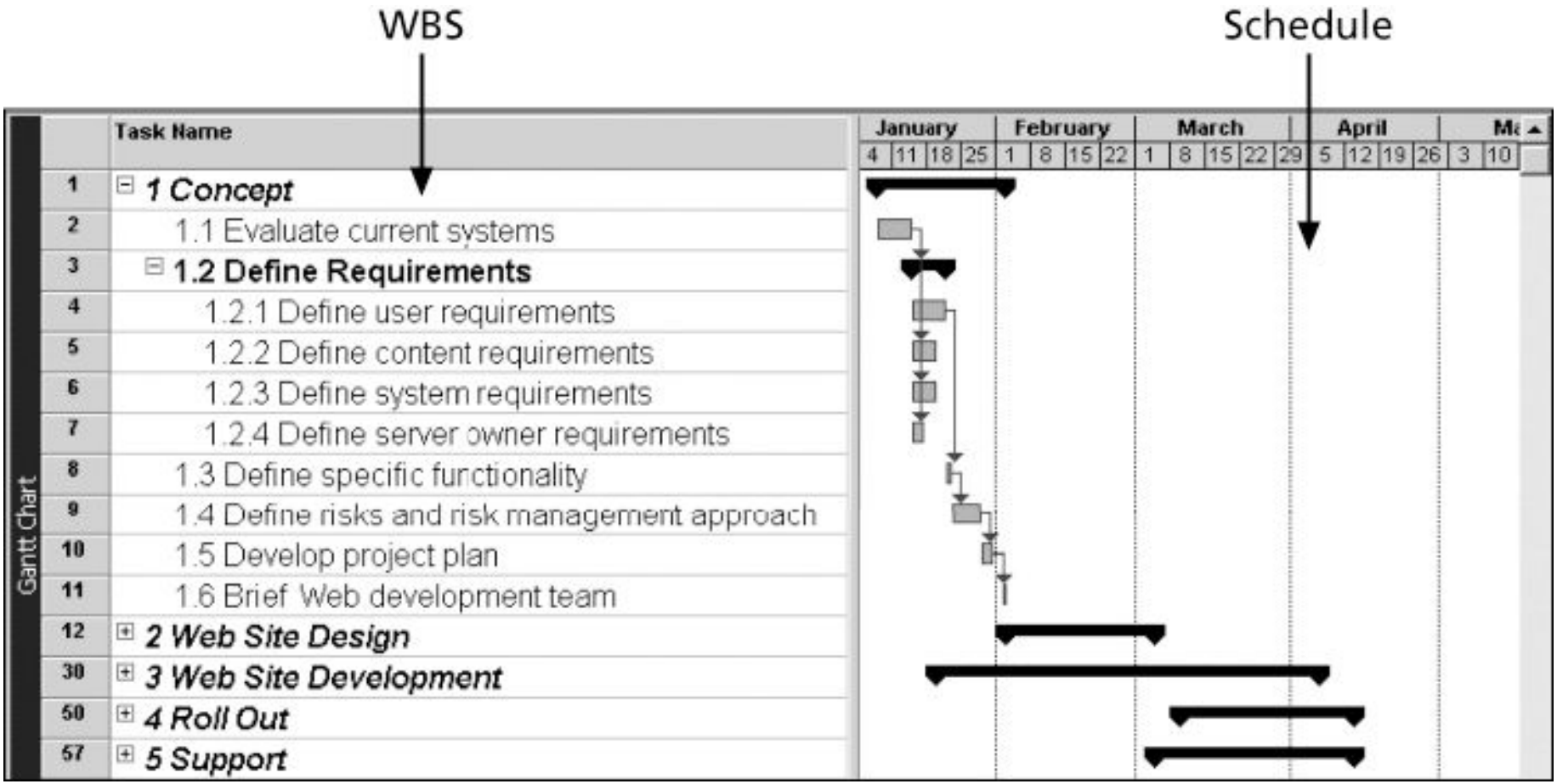


Figure 5-6. Intranet Gantt Chart Organized by Project Management Process Groups

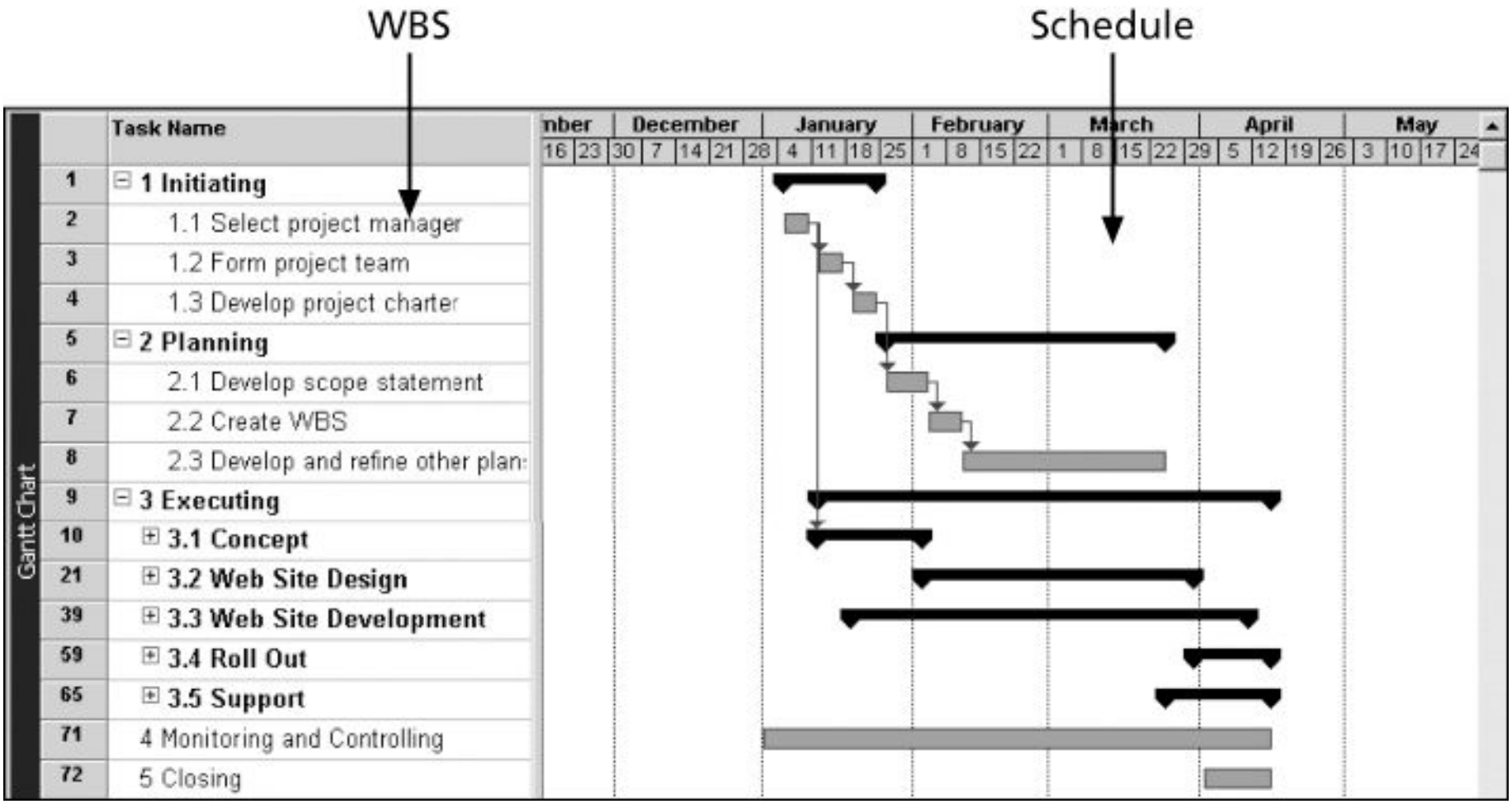


Table 5-4: Executing Tasks for JWD Consulting's WBS

3.0 Executing

- 3.1 Survey
- 3.2 User inputs
- 3.3 Intranet site content
  - 3.3.1 Templates and Tools
  - 3.3.2 Articles
  - 3.3.3 Links
  - 3.3.4 Ask the Expert
  - 3.3.5 User requests feature
- 3.4 Intranet site design
- 3.5 Intranet site construction
- 3.6 Site testing
- 3.7 Site promotion
- 3.8 Site roll out
- 3.9 Project benefits measurement

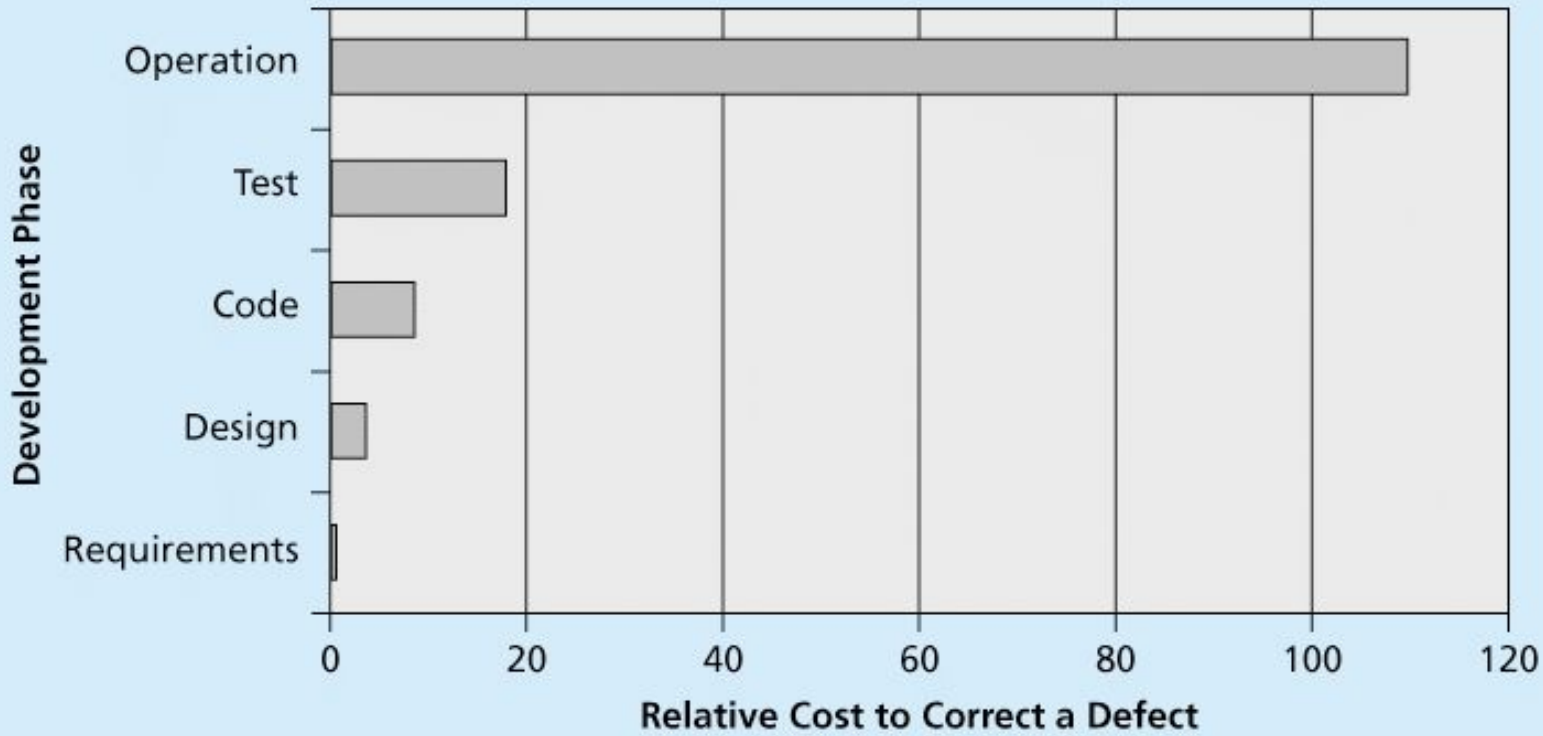
Approaches to Developing WBSs

- Using guidelines: Some organizations, like the DOD, provide guidelines for preparing WBSs
- The **analogy approach**: Review WBSs of similar projects and tailor to your project
- The **top-down approach**: Start with the largest items of the project and break them down
- The **bottom-up approach**: Start with the specific tasks and roll them up
- Mind-mapping approach: **Mind mapping** is a technique that uses branches radiating out from a core idea to structure thoughts and ideas

Collecting Requirements

- For some IT projects, it is helpful to divide requirements development into categories called elicitation, analysis, specification, and validation
- It is important to use an iterative approach to defining requirements since they are often unclear early in a project

Figure 5-2. Relative Cost to Correct a Software Requirement Defect



Source: Robert B. Grady, "An Economic Release Decision Model: Insights into Software Project Management." *Proceedings of the Applications of Software Measurement Conference* (Orange Park, FL: Software Quality Engineering, 1999), pp. 227–239.

Methods for Collecting Requirements

- Interviewing
- Focus groups and facilitated workshops
- Using group creativity and decision-making techniques
- Questionnaires and surveys
- Observation
- Prototyping
- **Benchmarking**, or generating ideas by comparing specific project practices or product characteristics to those of other projects or products inside or outside the performing

Statistics on Requirements for Software Projects (2011 Survey)*

- Eighty-eight percent of the software projects involved enhancing existing products instead of creating new ones
- Eighty-six percent of respondents said that customer satisfaction was the most important metric for measuring the success of development projects
- Eighty-three percent of software development teams still use Microsoft Office applications such as Word and Excel as their main tools to communicate requirements

*John Simpson, "2011: The State of Requirements Management" (2011).

What Went Right?

- Genesys Telecommunications Laboratories uses Accept software, a product planning and innovation management application and winner of the Excellence in Product Management Award from 2006–2008
- Accept helps them instill a consistent, repeatable, and predictable process for new product definition and development
- They can define what information comprises a requirement and enforce discipline around that process

Requirements Traceability Matrix

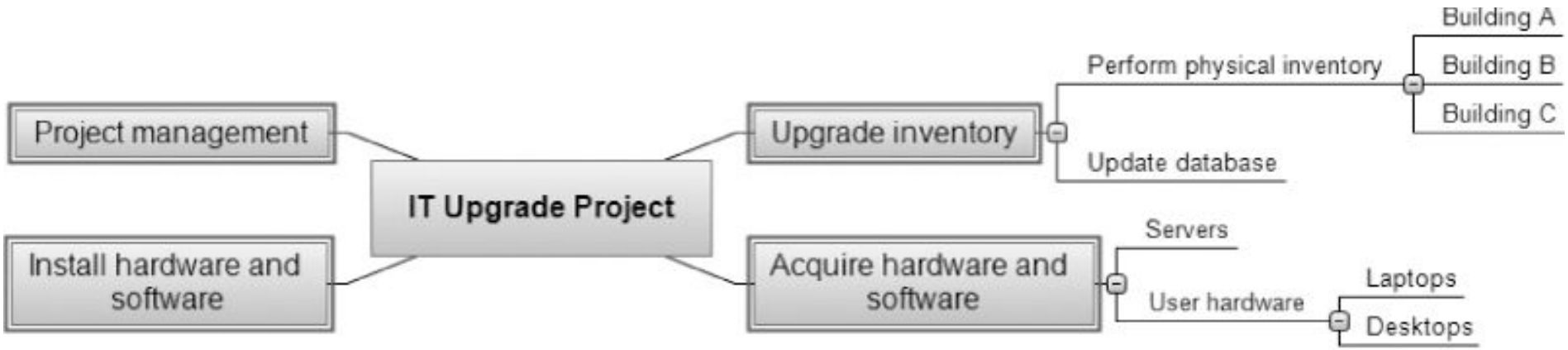
- A **requirements traceability matrix (RTM)** is a table that lists requirements, various attributes of each requirement, and the status of the requirements to ensure that all requirements are addressed
- Table 5-1. Sample entry in an RTM

Requirement No.	Name	Category	Source	Status
R32	Laptop memory	Hardware	Project charter and corporate laptop specifications	Complete. Laptops ordered meet requirement by having 4GB of memory.

Defining Scope

- **Project scope statements** should include at least a product scope description, product user acceptance criteria, and detailed information on all project deliverables. It is also helpful to document other scope-related information, such as the project boundaries, constraints, and assumptions. The project scope statement should also reference supporting documents, such as product specifications
- As time progresses, the scope of a project should become more clear and specific

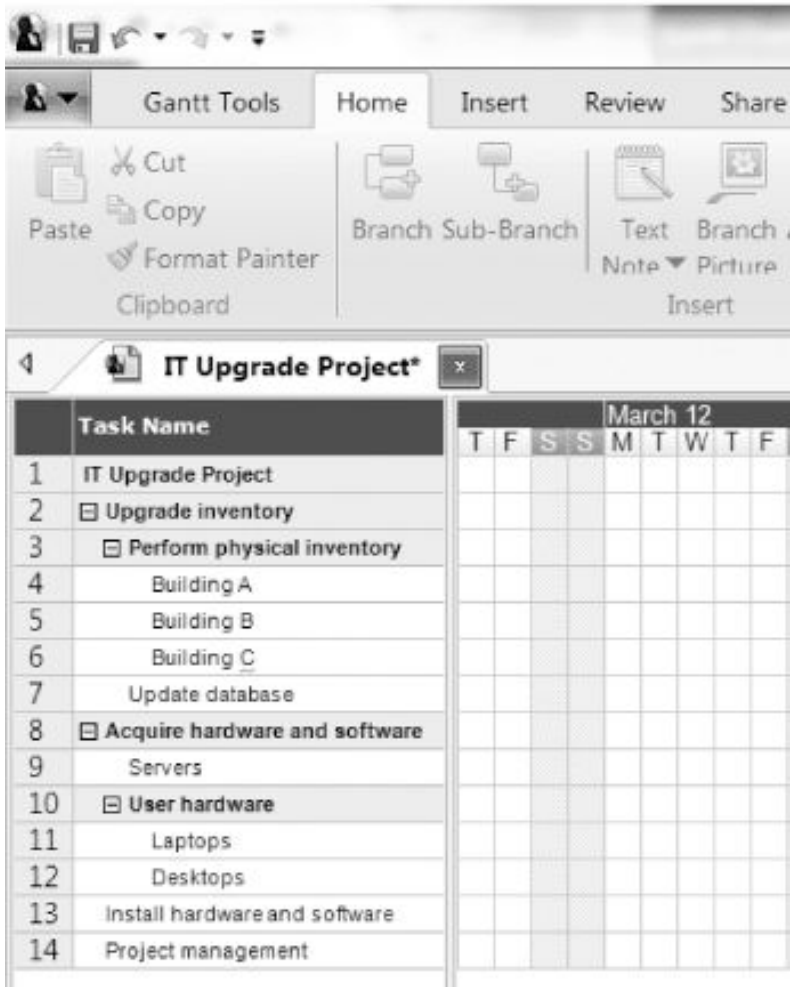
Figure 5-7. Sample Mind-Mapping Approach for Creating a WBS



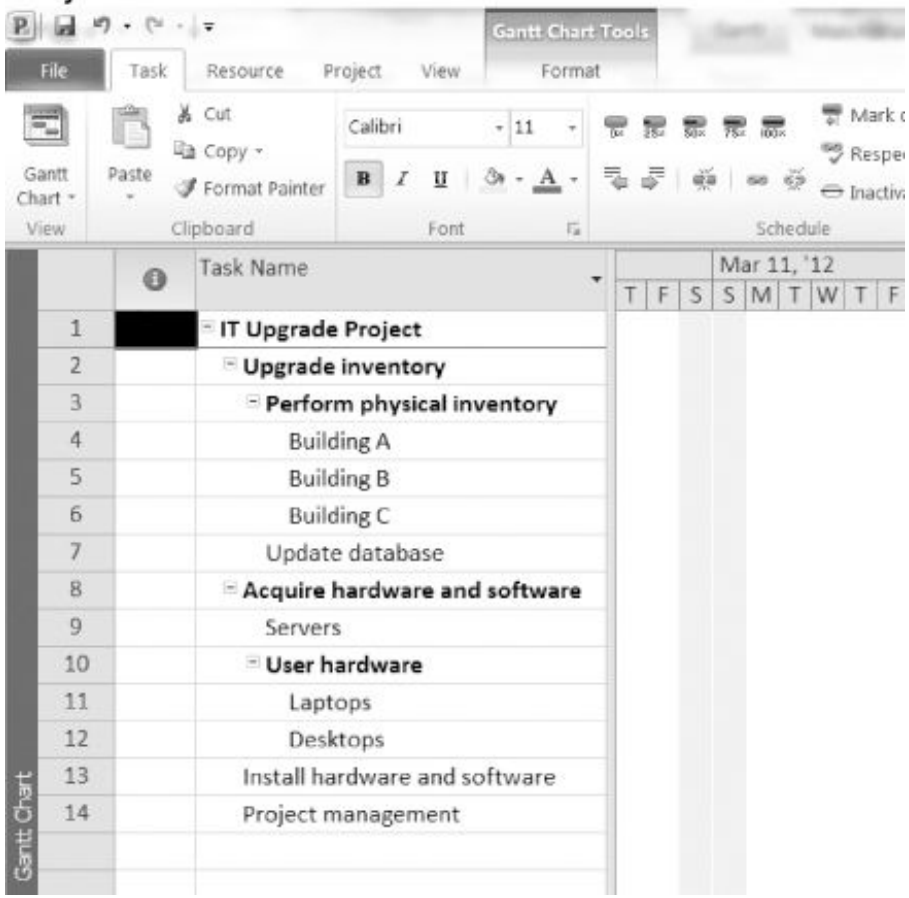
Source: MatchWare's MindView 4 Business Edition

Figure 5-8. Gantt Charts With WBS Generated From a Mind Map

MindView 4.0 Gantt Chart



Project 2010 Gantt Chart



The WBS Dictionary and Scope Baseline

- Many WBS tasks are vague and must be explained more so people know what to do and can estimate how long it will take and what it will cost to do the work
- A **WBS dictionary** is a document that describes detailed information about each WBS item

Table 5-5. Sample WBS Dictionary Entry

WBS Dictionary Entry March 20
Project Title: Information Technology (IT) Upgrade Project
WBS Item Number: 2.2
WBS Item Name: Update Database
Description: The IT department maintains an online database of hardware and software on the corporate intranet. However, we need to make sure that we know exactly what hardware and software employees are currently using and if they have any unique needs before we decide what to order for the upgrade. This task will involve reviewing information from the current database, producing reports that list each department's employees and location, and updating the data after performing the physical inventory and receiving inputs from department managers. Our project sponsor will send a notice to all department managers to communicate the importance of this project and this particular task. In addition to general hardware and software upgrades, the project sponsors will ask the department managers to provide information for any unique requirements they might have that could affect the upgrades. This task also includes updating the inventory data for network hardware and software. After updating the inventory database, we will send an e-mail to each department manager to verify the information and make changes online as needed. Department managers will be responsible for ensuring that their people are available and cooperative during the physical inventory. Completing this task is dependent on WBS Item Number 2.1, Perform Physical Inventory, and must precede WBS Item Number 3.0, Acquire Hardware and Software.

Advice for Creating a WBS and WBS Dictionary

- A unit of work should appear at only one place in the WBS.
- The work content of a WBS item is the sum of the WBS items below it
- A WBS item is the responsibility of only one individual, even though many people may be working on it
- The WBS must be consistent with the way in which work is actually going to be performed; it should serve the project team first, and other purposes only if practical

Advice for Creating a WBS and WBS Dictionary (cont'd)

- Project team members should be involved in developing the WBS to ensure consistency and buy-in
- Each WBS item must be documented in a WBS dictionary to ensure accurate understanding of the scope of work included and not included in that item
- The WBS must be a flexible tool to accommodate inevitable changes while properly maintaining control of the work content in the project according to the scope statement

What Went Wrong?

- A project scope that is too broad and grandiose can cause severe problems
  - Scope creep and an overemphasis on technology for technology's sake resulted in the bankruptcy of a large pharmaceutical firm, Texas-based FoxMeyer Drug
  - In 2001, McDonald's fast-food chain initiated a project to create an intranet that would connect its headquarters with all of its restaurants to provide detailed operational information in real time. After spending $170 million on consultants and initial implementation planning, McDonald's

Validating Scope

- It is very difficult to create a good scope statement and WBS for a project
- It is even more difficult to verify project scope and minimize scope changes
- **Scope validation** involves formal acceptance of the completed project deliverables
- Acceptance is often achieved by a customer inspection and then sign-off on key deliverables

Global Issues

- Many countries have had difficulties controlling the scope of large projects, especially those that involve advanced technologies and many different users
- For example, the state government of Victoria, Australia, has a Web site for its public transportation smart card at www.myki.com.au.
- There were many problems in developing and implementing the smart card

Controlling Scope

- Scope control involves controlling changes to the project scope
- Goals of scope control are to
  - influence the factors that cause scope changes
  - assure changes are processed according to procedures developed as part of integrated change control, and
  - manage changes when they occur
- **Variance** is the difference between planned and actual performance

Best Practices for Avoiding Scope Problems

1. Keep the scope realistic. Don't make projects so large that they can't be completed. Break large projects down into a series of smaller ones
2. Involve users in project scope management. Assign key users to the project team and give them ownership of requirements definition and scope verification
3. Use off-the-shelf hardware and software whenever possible. Many IT people enjoy using the latest and greatest technology, but business needs, not technology trends, must take priority
4. Follow good project management processes. As described in this chapter and others, there are well-defined processes for managing project scope and others aspects of projects

Suggestions for Improving User Input

- Develop a good project selection process and insist that sponsors are from the user organization
- Have users on the project team in important roles
- Have regular meetings with defined agendas, and have users sign off on key deliverables presented at meetings
- Deliver something to users and sponsors on a regular basis
- Don’t promise to deliver when you know you can’t

Suggestions for Reducing Incomplete and Changing Requirements

- Develop and follow a requirements management process
- Use techniques such as prototyping, use case modeling, and JAD to get more user involvement
- Put requirements in writing and keep them current
- Create a requirements management database for documenting and controlling requirements

Suggestions for Reducing Incomplete and Changing Requirements (cont'd)

- Provide adequate testing and conduct testing throughout the project life cycle
- Review changes from a systems perspective
- Emphasize completion dates to help focus on what's most important

Using Software to Assist in Project Scope Management

- Word-processing software helps create several scope-related documents
- Spreadsheets help to perform financial calculations, weighed scoring models, and develop charts and graphs
- Communication software like e-mail and the Web help clarify and communicate scope information
- Project management software helps in creating a WBS, the basis for tasks on a Gantt chart

You are the project manager for IT company who will built a hotel management systems for a client ABC. Assuming you have a project charter initiated, clearly outline a WBS for this project for 1 yea

Level 1: Project Title

Hotel Management System Development for Client ABC

Level 2: Project Phases

1. **Project Initiation**
2. **Requirements Analysis**
3. **System Design**
4. **Development**
5. **Testing**
6. **Deployment and Training**
7. **Project Closure**

Level 3: Tasks and Subtasks

1. Project Initiation

1.1 Define project scope
1.2 Identify stakeholders
1.3 Finalize project charter
1.4 Set up project management tools
1.5 Conduct kick-off meeting

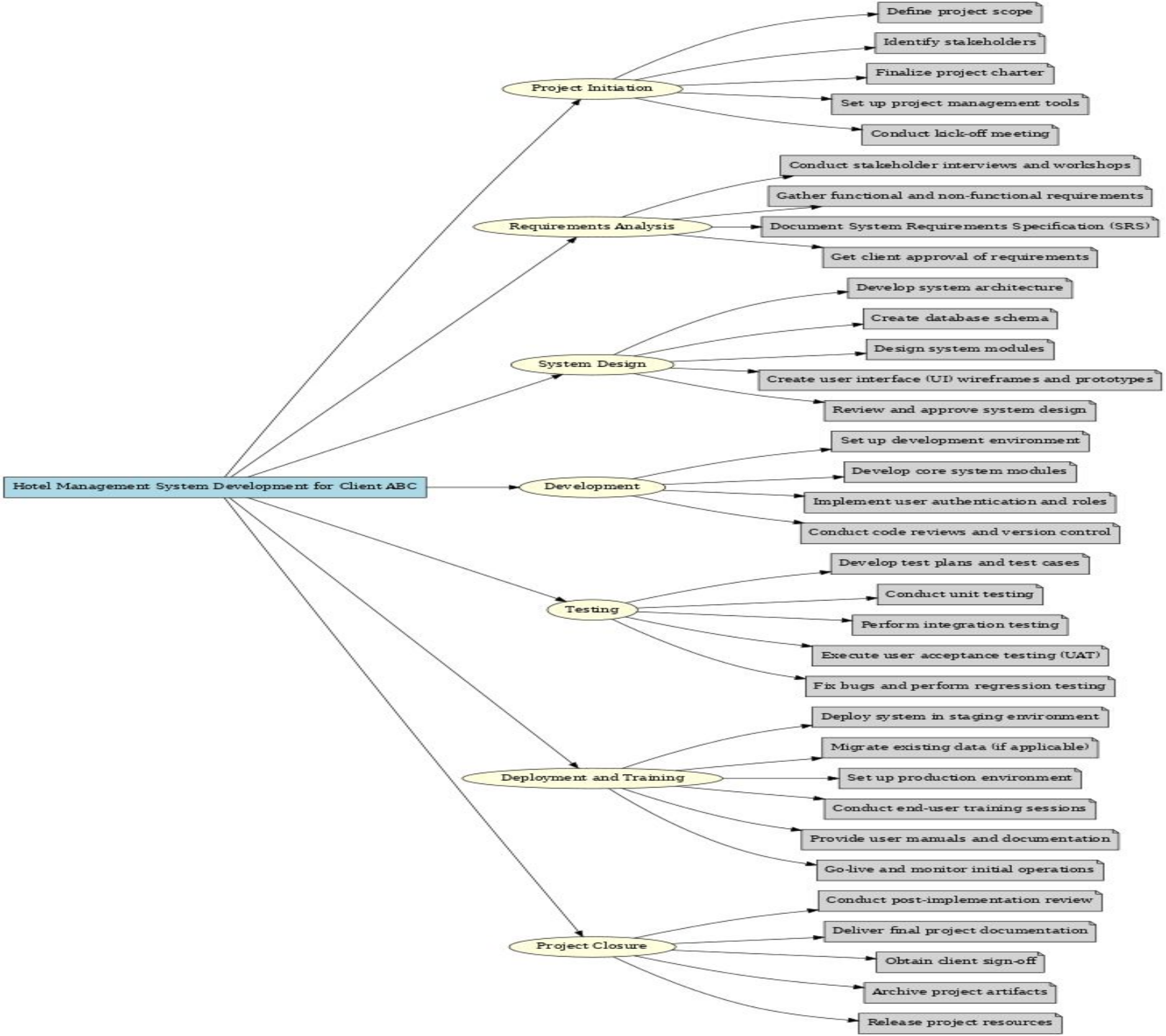
2. Requirements Analysis

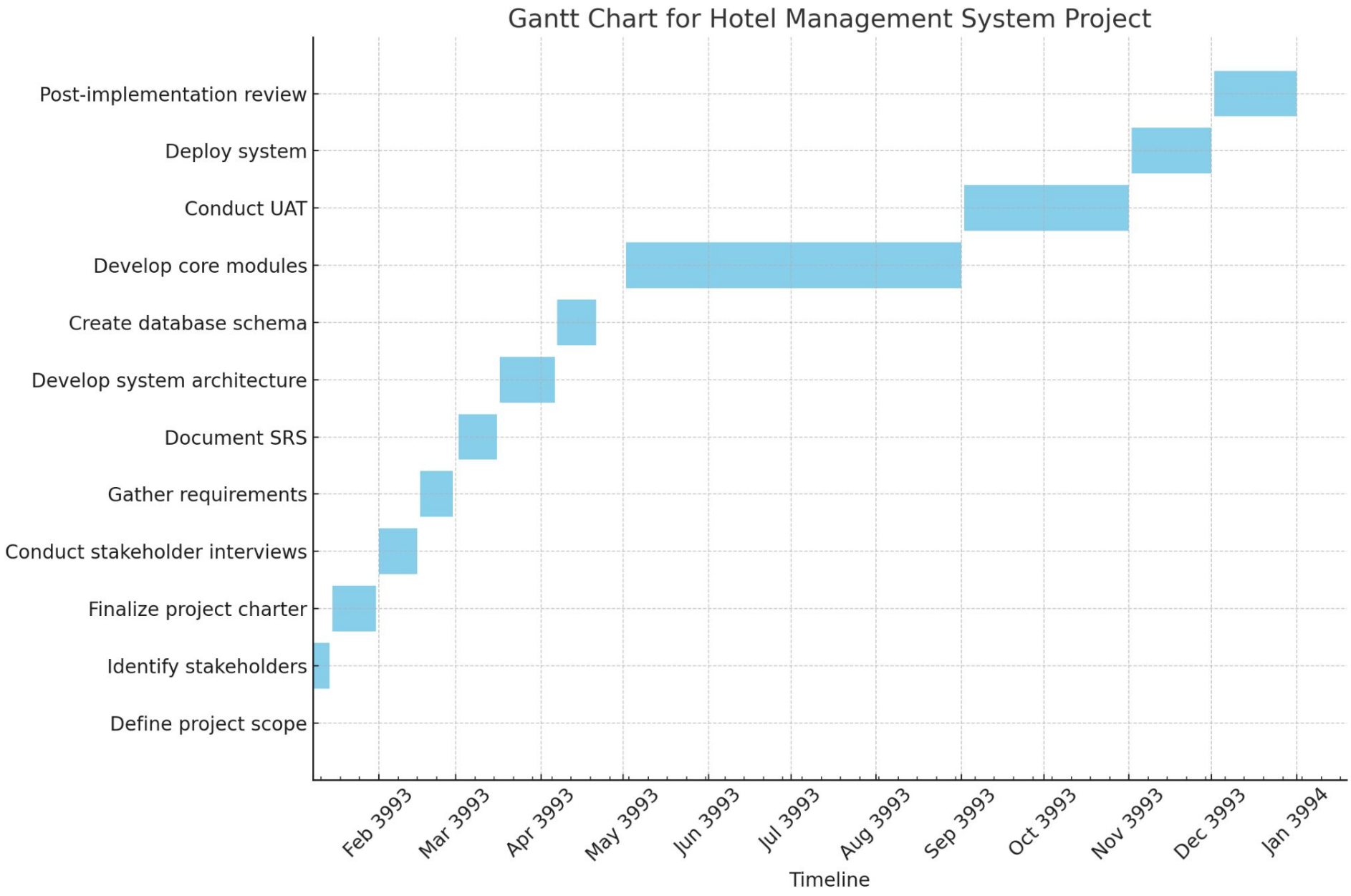
2.1 Conduct stakeholder interviews and workshops
2.2 Gather functional and non-functional requirements
2.3 Document System Requirements Specification (SRS)
2.4 Get client approval of requirements

3. System Design

Timeline Summary

Phase	Duration	Timeline
Project Initiation	1 month	Month 1
Requirements Analysis	1.5 months	Month 2 - Mid Month 3
System Design	1.5 months	Mid Month 3 - Month 4
Development	4 months	Month 5 - Month 8
Testing	2 months	Month 9 - Month 10
Deployment and Training	1 month	Month 11
Project Closure	1 month	Month 12


Hotel Management System Development for Client ABC
Project Initiation
Define project scope
Identify stakeholders
Finalize project charter
Set up project management tools
Conduct kick-off meeting
Requirements Analysis
Conduct stakeholder interviews and workshops
Gather functional and non-functional requirements
Document System Requirements Specification (SRS)
Get client approval of requirements
System Design
Develop system architecture
Create database schema
Design system modules
Create user interface (UI) wireframes and prototypes
Review and approve system design
Development
Set up development environment
Develop core system modules
Implement user authentication and roles
Conduct code reviews and version control
Testing
Develop test plans and test cases
Conduct unit testing
Perform integration testing
Execute user acceptance testing (UAT)
Fix bugs and perform regression testing
Deployment and Training
Deploy system in staging environment
Migrate existing data (if applicable)
Set up production environment
Conduct end-user training sessions
Provide user manuals and documentation
Go-live and monitor initial operations
Project Closure
Conduct post-implementation review
Deliver final project documentation
Obtain client sign-off
Archive project artifacts
Release project resources


Gantt Chart for Hotel Management System Project
Post-implementation review
Deploy system
Conduct UAT
Develop core modules
Create database schema
Develop system architecture
Document SRS
Gather requirements
Conduct stakeholder interviews
Finalize project charter
Identify stakeholders
Define project scope
Feb 3993
Mar 3993
Apr 3993
May 3993
Jun 3993
Jul 3993
Aug 3993
Sep 3993
Oct 3993
Nov 3993
Dec 3993
Jan 3994
Timeline

Chapter Summary

- Project scope management includes the processes required to ensure that the project addresses all the work required, and only the work required, to complete the project successfully
- Main processes include
  - Define scope management
  - Collect requirements
  - Define scope
  - Create WBS
  - Validate scope
  - Control scope